

The Impact of Business Training on Firm Revenue: Evidence from Panel Data

Jesugnon David Janvier Koumagnon

October 5, 2025

Abstract

Businesses operate with the primary goal of generating profit while efficiently managing resources, and effective training of managers and employees is often critical to achieving this objective. This study investigates the impact of a targeted training program on firm performance, focusing on how capacity-building influences revenue, sales, employment, and productivity.

We use a panel of firm-level monthly data from 2010 to 2019, combined with daily sales and baseline firm information, and employ a Difference-in-Differences approach with two-way fixed effects, complemented by robustness checks and a Regression Discontinuity design around the 100-employee eligibility threshold. Our findings indicate that trained firms experienced substantial gains in financial outcomes, with revenue increasing by 38%, sales by 22%, and revenue per worker by 38%, while employment remained largely unchanged. These results suggest that training interventions can significantly enhance operational efficiency and firm competitiveness without necessarily expanding the workforce.

1 Introduction

Business organizations face persistent pressure to improve performance while efficiently deploying limited resources. This challenge has motivated policymakers to design interventions aimed at strengthening managerial capabilities, particularly among small and medium enterprises that often lack access to formal business training. Despite widespread implementation of such programs, fundamental questions remain about their effectiveness and the mechanisms through which they influence firm behavior.

[McKenzie and Woodruff \(2014\)](#) synthesized evidence from multiple randomized experiments, documenting considerable heterogeneity in program impacts in different contexts and designs. Although some interventions successfully improved business practices and firm survival, others generated modest effects on sales, profits, or employment. In addition, [Bloom et al. \(2013\)](#) provided intensive management consulting to Indian textile manufacturers and documented productivity increases averaging 17 percent within the first year. However, research focusing specifically on training programs for small business owners has produced more mixed results. [Bruhn et al. \(2018\)](#) evaluated a business training program in Mexico through a randomized trial, finding positive short-term effects on business practices but limited impacts on sales or profits after two years.

The paper examines a business training program implemented for small and medium enterprises with distinctive institutional features that enable credible causal inference. The program, which began in January 2013, offered management training to firms with 100 or fewer employees. This eligibility threshold creates a natural experiment in which firms just below the cutoff gained access to training, while similar firms just above the threshold remained ineligible. Among eligible firms, participation was voluntary, introducing variation in treatment receipt that we leverage through instrumental variables estimation.

Our research question is what causal effects does the training program generate on firm employment, revenue, sales, and labor productivity? The primary specification implements a difference-in-differences estimator with firm and time fixed effects, comparing outcomes for participating and nonparticipating firms before and after the program's 2013 launch. We evaluated the validity of the parallel trends assumption through event study analysis and implemented a regression discontinuity design that exploited the sharp discontinuity in eligibility in 100 employees.

2 Data

2.1 Processing and Cleaning

This study relies on three complementary firm-level datasets: (i) firm information, (ii) daily aggregate sales, and (iii) monthly operational data spanning the period from January 1, 2010, to December 31, 2019. The datasets were preprocessed, harmonized, and merged to construct a balanced panel suitable for causal inference on the impact of a training program on firm outcomes.

The first dataset, firm information, includes 498 unique firms and contains identifiers, firm names, and sector classifications. This dataset required no cleaning as it contained no duplicates, ensuring that each firm could be uniquely linked to the other datasets.

The second dataset, aggregate daily sales, includes 56,646 daily observations for the firms over the ten-year period. During preprocessing, it was necessary to standardize firm identifiers and dates. Some files recorded dates using the day–month–year format, while others used the month–day–year format. To address this inconsistency, a custom parsing procedure was implemented to correctly interpret and convert all date entries into a uniform format. Firm identifiers were cleaned by removing leading and trailing whitespaces, converting all characters to uppercase, and eliminating internal spaces. Only the standardized alphanumeric pattern of identifiers was retained to ensure consistent linkage across sources. These adjustments were essential to prevent merging errors during data integration.

The third dataset, monthly firm-level data, contains daily records on key operational measures such as employment, wage bills, revenues, and program participation. These monthly data were initially stored across multiple files, which were combined into a single panel. Numeric variables were converted to consistent formats, firm identifiers were standardized using the same procedure as in the sales data, and date variables were harmonized across differing formats. This cleaning ensured that all monthly observations were correctly aligned with the aggregate sales dataset.

Once standardized, the daily sales data were merged with the monthly dataset by firm identifier and date. Subsequently, firm information was appended to include firm names and sector classifications. This three-way merge produced a unified dataset containing firm characteristics, financial performance, and program adoption. The merged data were then aggregated to the firm–month level by computing monthly averages for employment, revenues, sales, and wage bills. Program participation was recorded as one if a firm participated at any point within the month, ensuring that partial-month participation was accurately captured. The number of daily observations contributing to each monthly record was also stored to monitor data completeness.

From the aggregated dataset, additional variables were created to support causal analysis. Temporal indicators for year and month were derived to structure the panel. Firms that enrolled in the training program after 2013 were classified as treated, while others served as controls. A post-program indicator was defined for periods from 2013 onward, and an interaction term between treatment status and the post-program period was constructed to identify the treatment effect in difference-in-differences spec-

ifications. To facilitate event-study analysis, the number of months elapsed since January 2013 was calculated.

Eligibility for the program was determined using firms’ average employment levels over the pre-treatment period (2010–2012). A binary indicator was defined for firms with fewer than or equal to 100 employees, corresponding to the program’s eligibility threshold. A continuous measure of distance from the threshold was also computed to capture how close firms were to the eligibility cutoff. In addition, an indicator identified firms above this limit.

To ensure interpretability and mitigate skewness in the data, key variables were expressed in logarithmic form. Log transformations were applied to employment, revenues, sales, and wage bills. Productivity was proxied by the ratio of revenues to employment, and its logarithmic transformation was used for empirical estimation.

The resulting monthly panel contains one observation per firm per month and exhibits excellent coverage. Of the 498 firms, 478 (96 percent) have baseline data prior to 2013, while the remaining 20 firms appear to have entered the dataset after the pre-treatment period. The number of active firms per year remained stable, ranging from 473 to 476. Treated and control firms display comparable observation lengths, with a mean of 112 months and a median of 118 months, providing a consistent basis for longitudinal analysis.

2.2 Sample Coverage and Baseline Information

Table 1 reports the distribution of firms with baseline data prior to 2013. Out of 498 firms in total, 478 (approximately 96 percent) have pre-treatment observations. Only 20 firms lack baseline information, suggesting that they likely entered the market after the program was introduced or began reporting later. This high coverage rate ensures that the identification of treatment effects is not biased by differential data availability.

Table 1: Baseline Data Coverage Before 2013

	Number of Firms	Share (%)
Firms with baseline data	478	95.98
Firms without baseline data	20	4.02
Total	498	100

The high level of baseline coverage supports the internal validity of the empirical strategy, as the majority of firms can be compared over a sufficiently long pre-treatment period.

2.3 Temporal Distribution of Firms

Table 2 summarizes the number of active firms observed in each year from 2010 to 2019. The annual number of firms remains remarkably stable, fluctuating between 473 and 476. This temporal consistency

suggests limited attrition and strengthens the reliability of longitudinal analyses.

Table 2: Number of Active Firms per Year

Year	Number of Firms
2010	473
2011	473
2012	475
2013	474
2014	476
2015	473
2016	476
2017	473
2018	475
2019	475

The nearly constant firm counts over time indicate that the panel is well-balanced and that sample exits or entries are minimal. Such stability reduces concerns about selection bias driven by firm turnover.

2.4 Treatment and Control Groups

The program targeted small firms with no more than 100 employees. Among the 498 firms, 181 enrolled in the program after its introduction in 2013, while 317 did not. Table 3 reports descriptive statistics for treated and untreated firms, including the average number of monthly observations per firm. The treated and control firms display nearly identical observation lengths, with both groups averaging 112 months of data. The minimum number of observed months is slightly higher among treated firms, suggesting that they tend to be present in the dataset for longer continuous periods. This reinforces the reliability of subsequent causal analyses, as differences between treated and control firms are unlikely to stem from data availability.

Table 3: Treatment and Control Group Summary

Group	Number of Firms	Mean Months	Median Months	Minimum Months
Control	317	112	118	7
Treated	181	112	118	17

2.5 Pre-Treatment Differences Across Firms

To provide a detailed view of the characteristics of the firm prior to the introduction of the program in January 2013, Tables 4 and 5 present descriptive statistics over the period 2010-2012. The first table

compares firms that would later adopt the program (treated) to those that did not (control), while the second compares firms by their ex-ante eligibility status based on employment size. Table 4 reveals that Firms that eventually participated in the program were, on average, considerably smaller and less capitalized than non-treated firms. Their average employment level (63 workers) is roughly half that of the control group (114 workers), and their mean revenues and wage bills are proportionally lower. This pattern is consistent with the program’s design, which specifically targeted smaller firms with limited human and financial resources. The higher standard deviations among treated firms also indicate greater heterogeneity in firm size and performance among program participants.

Table 4: Descriptive Statistics by Treatment Status (2010–2012)

Variable	Control	Treated
n_{firms}	304	174
n_{obs}	10,639	6,054
Mean employment	113.621	63.418
SD employment	33.805	25.049
Mean revenue	823,501.693	254,126.374
SD revenue	535,087.921	258,253.515
Mean sales	10,309.167	4,494.965
SD sales	4,664.018	2,878.512
Mean wage bill	35,862.933	17,098.585
SD wage bill	14,360.036	9,153.933

Table 5 compares the companies that were eligible for the program (those employing 100 workers or fewer) with those that were not. The differences in scale and performance are pronounced. Ineligible firms, typically larger, exhibit mean employment levels twice as high as eligible firms and earn, on average, more than four times their revenue. Their average wage bill and sales volumes follow the same pattern, reflecting both greater production capacity and higher levels of formality. In contrast, eligible firms are smaller, less diversified, and have a significantly lower dispersion in employment and revenue, suggesting a more homogeneous composition among smaller enterprises.

Table 5: Descriptive Statistics by Eligibility Status (2010–2012)

Variable	Ineligible (>100)	Eligible (≤ 100)
n_{firms}	228	250
n_{obs}	8,072	8,621
Mean employment	129.792	63.293
SD employment	19.735	21.815
Mean revenue	1,028,841.190	231,769.743
SD revenue	468,720.821	180,085.060
Mean sales	12,324.449	4,352.110
SD sales	3,544.925	2,303.232
Mean wage bill	42,011.728	16,929.849
SD wage bill	10,604.756	7,834.500

We conducted a t-test to check if there is a difference in revenue, sales and employment during the pre-training stage. The result of each test confirm that there is significant difference accross the groups before january 2013 In addition, the large gap in average employment between eligible and ineligible firms confirms that the 100-employee threshold is a relevant and binding criterion for program participation. The data also suggest that most of the treated firms fall within the eligibility group, reinforcing the credibility of using eligibility as a proxy for treatment exposure in robustness analyzes.

3 Empirical Strategy

Our empirical strategy combines several complementary approaches to estimate the causal effect of the training program on firm outcomes: (i) a difference-in-differences (DiD) specification with two-way fixed effects, (ii) an event-study formulation to assess dynamics and test the parallel trends assumption, (iii) placebo and balanced-panel robustness checks, and (iv) a regression discontinuity design (RDD) around the program eligibility cutoff.

3.1 Notation and variable definitions

Let i index firms and t index months. The main outcome variables used in the paper are the month-firm aggregates constructed in the data stage:

- $revenue_t$: monthly revenue (monthly average of daily revenue observations).
- $sales_t$: monthly sales (monthly average of daily sales observations).
- $employment_t$: monthly employment (monthly average daily employment).
- $wage_bill_t$: monthly wage bill (monthly average of daily wage bill).

To reduce skewness and to interpret effects approximately in percentage terms, we use logarithmic transformations. In the code we create:

$$\log revenue = \log(revenue_t + 1), \quad \log sales = \log(sales_t + 1)$$

$$\log employment = \log(employment_t + 1), \quad \log wage_bill = \log(wage_bill_t + 1).$$

The +1 adjustment avoids taking the logarithm of zero.

Treatment and timing variables are defined as:

- $treated$: an indicator that equals 1 for firms that ever participated in the program in the post-2013 period (constructed by detecting $adopt_t = 1$ for any month in ≥ 2013).
- $post_treatment$: an indicator equal to 1 for all months in year ≥ 2013 .
- $treated_post = treated \times post_treatment$: the DiD interaction capturing treated firms in the post-program period.
- $months_since_2013$: months elapsed since January 2013 (used to build event-study bins).

Eligibility variables (used in RDD and descriptive work) are computed from pre-treatment averages (2010–2012):

- $employment_baseline_i$: firm i 's average employment before 2013.

- $eligible_i$: indicator equal to 1 if $employment_baseline \leq 100$.
- $distance_from_threshold = employment_baseline - 100$: distance to the 100-employee eligibility cutoff.
- $above_threshold$: indicator equal to 1 if $employment_baseline > 100$.

3.2 Main difference-in-differences (DiD) specification

The baseline causal specification is a standard two-way fixed effects DiD that estimates the average effect of being a treated firm after the program began. Two-way fixed effects control for unobserved heterogeneity across firms and for common shocks that affect all firms in a given month (e.g., macro shocks, seasonality). Because program participation is not randomized, these fixed effects help reduce bias from time-invariant confounders and common temporal shocks. The DiD design is appropriate because some eligible firms enrolled and others did not.. The estimating equation is:

$$y_{it} = \alpha_i + \delta_t + \beta \text{treated_post}_{it} + \varepsilon_{it}, \quad (1)$$

where:

- y_{it} is an outcome in logs (e.g., $\log revenue$, $\log sales$, $\log employment$),
- α_i are firm fixed effects that absorb all time-invariant firm characteristics,
- δ_t are time fixed effects (implemented as year×month fixed effects in the code), and
- ε_{it} is an idiosyncratic error term. The coefficient β measures the average treated vs control difference in the post period relative to the pre period.

3.3 Event-study specification: dynamics and parallel trends

To examine dynamic effects and test the parallel trends assumption, we estimate an event-study specification that replaces the single DiD indicator with a set of relative time indicators. The event study provides (i) a visual and statistical test of the parallel-trends assumption by checking whether pre-treatment γ_k coefficients are statistically indistinguishable from zero, and (ii) an estimate of how effects evolve after treatment. If pre-treatment coefficients are flat, this supports the identifying parallel trends assumption required for the DiD estimator. Let k index discrete time bins relative to program start. The event-study equation estimated in the code is:

$$y_{it} = \alpha_i + \delta_t + \sum_{k \neq k_0} \gamma_k D_{i,t}^{(k)} + \varepsilon_{it}, \quad (2)$$

where $D_{i,t}^{(k)}$ equals 1 if observation (i,t) falls in relative time bin k for treated firms and 0 otherwise. The omitted (reference) bin k_0 in the code is the window $[-6, -3]$ months before January 2013. In the implementation the code collapses months into broader bins (e.g., ≤ -18 months, -15 to -12 , etc.) to ensure sufficient observations per bin.

3.4 Robustness and placebo checks

We perform a set of robustness checks to probe alternative explanations and mechanical biases. The placebo assesses whether pre-program movements could be mistaken for treatment effects, and the balanced panel ensures that results are not driven by a small group of firms with incomplete records. Both checks increase confidence in causal interpretation when results are consistent with the baseline.

Placebo test. A placebo DiD is estimated by pretending the program started in 2011. Concretely, we restrict the sample to pre-2013 observations and define a fake post indicator equal to 1 for years ≥ 2011 . The regression is:

$$y_{it} = \alpha_i + \delta_t + \beta^{\text{placebo}} \text{fake_treated_post}_{it} + \varepsilon_{it}.$$

A statistically insignificant β^{placebo} provides evidence against spurious pre-existing trends driving the main results.

We restrict to firms with a complete 10-year record (120 months) and re-estimate the main DiD. This addresses concerns that differential entry/exit might bias estimates. The balanced sample is constructed in the code by selecting firms with exactly 120 monthly observations and re-running the TWFE DiD.

3.5 Regression discontinuity design (RDD)

The program has an eligibility rule based on pre-treatment employment: firms with baseline employment ≤ 100 are eligible. To exploit this threshold more directly, we implement a cross-sectional RDD using the baseline employment measure. The RDD isolates quasi-random variation in treatment assignment for firms close to the eligibility cutoff, providing a local causal estimate that is robust to many forms of selection that are smooth in the running variable. The RDD estimating equation is:

$$y_i = \tau \cdot \mathbf{1}\{\text{distance_from_threshold}_i \geq 0\} + f(\text{distance_from_threshold}_i) + u_i, \quad (3)$$

where:

- y_i is a post-treatment outcome averaged over the post period (in the code we filter to year ≥ 2013 and use monthly observations but restrict to firms within a symmetric window around the threshold).
- The function $f(\cdot)$ denotes a smooth function of the running variable (distance from 100 employees), which in practice is approximated using local polynomials.

- The parameter τ captures the local treatment effect at the threshold.

Throughout all specifications we report heteroskedasticity-robust standard errors clustered at the firm level. Clustering addresses serial correlation in the residuals within firms and cross-section dependence that might otherwise understate standard errors.

3.6 Summary of identification assumptions

Identification of the average treatment effect from the DiD (equation 1) requires that, conditional on firm fixed effects and time fixed effects, treated and control firms would have followed parallel trends in the absence of the program. The event study (equation 2) provides an empirical test of this assumption by checking for pre-treatment differences. The balanced panel and placebo exercises probe threats from differential attrition or pre-existing trends. The RDD (equation 3) relies on continuity of potential outcomes around the eligibility cutoff and identifies a local effect for firms close to the threshold; it therefore complements the DiD by exploiting quasi-random assignment near the eligibility rule.

In the next section we present the estimation results for the outcomes described above, beginning with the two-way fixed effects DiD estimates and following with event-study plots, robustness checks (placebo and balanced panel), and the RDD results.

4 Empirical Results

4.1 Main Difference-in-Differences Estimates

Table 6 presents the baseline differences in differences estimates of the program’s effects on firm outcomes, including employment, revenue, sales and productivity. All regressions include firm-fixed effects to control for time-invariant heterogeneity across firms and time-fixed effects to capture common macroeconomic shocks. Standard errors are clustered at the firm level to account for serial correlation.

The interaction term $Treated \times Post$ captures the average change in the results of the treated firms after the introduction of the program, relative to the control group. The estimates do not reveal a statistically significant impact on employment: the coefficient is close to zero (-0.002) and insignificant, suggesting that participation in the program did not alter the employment levels of the firms during the study period.

In contrast, the coefficients for revenue, sales and productivity are positive and highly significant. On average, program participation is associated with approximately a **37.6% increase in revenue**, a **21.8% increase in sales**, and a **37.9% increase in productivity** after adoption.

The lack of a corresponding increase in employment, combined with substantial gains in financial and productivity measures, implies that firms improved their efficiency rather than expanded their workforce. In other words, the program appears to have enabled firms to generate higher output and revenues using their existing resources, potentially through better managerial practices, process optimization, or technology adoption. The inclusion of firm and time fixed effects ensures that these findings are not driven by unobserved firm-specific characteristics or aggregate shocks, supporting the internal validity of the identification strategy.

Table 6: Difference-in-Differences Results: Training Program Effects

	<i>Dependent variable:</i>			
	Employment	Revenue	Sales	Productivity
	(1)	(2)	(3)	(4)
Treated $\times$ Post	−0.002 (0.001)	0.376*** (0.028)	0.218*** (0.018)	0.379*** (0.027)
Firm FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	52,767	52,743	52,706	50,773
R ²	0.00003	0.078	0.054	0.076
Adjusted R ²	−0.009	0.069	0.046	0.067

Note: *p<0.1; **p<0.05; ***p<0.01. RSE clustered at the firm level.

4.2 Dynamic Effects: Event-Study Evidence

To explore the timing of the effects, Figure 1 reports event-study estimates of the yearly treatment effects relative to the pre-treatment period. The coefficients for the pre-treatment years fluctuate around zero and are statistically insignificant, confirming that treated and control firms followed parallel trends prior to program implementation. This finding reinforces the validity of the Difference-in-Differences identification assumption.

After the program's introduction, the estimated coefficients for revenue, sales, and productivity rise sharply and remain persistently positive over time. This pattern indicates that program effects emerged shortly after adoption and continued to strengthen in subsequent years. The persistence of these gains suggests that the program's impact was not transitory but rather reflected lasting improvements in firm practices or capabilities. The absence of pre-trends and the sustained post-treatment increases together provide strong empirical support for a causal interpretation of the

program's effects on firm performance.

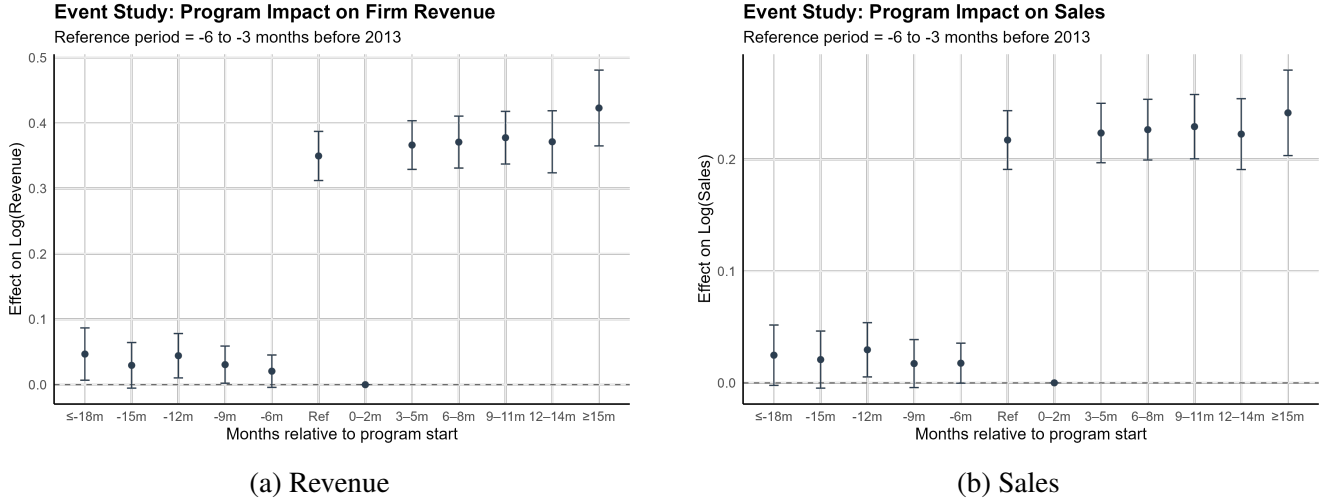


Figure 1: Event study graphs for revenue and sales

4.3 Robustness Checks

To ensure the reliability of the baseline Difference-in-Differences results, a series of robustness tests were conducted. Table 7 presents three complementary specifications: (i) the main model, (ii) a balanced panel restricted to firms continuously observed from 2010 to 2019, and (iii) a placebo test introducing a fictitious treatment in 2011, two years before the program started.

The balanced panel analysis provides a stringent robustness check by restricting attention to firms with complete pre- and post-treatment observations. This eliminates concerns that the main estimates might be driven by sample entry or attrition over time. The estimated effect remains positive and statistically significant at the 1% level, with an estimated coefficient of 0.506. This corresponds to an approximate **50.6% increase in firm revenue**, which is even larger than the baseline estimate of 37.6%. The consistency in sign and magnitude across specifications demonstrates that the observed revenue gains are not attributable to sample composition or selective participation.

The placebo test introduces a simulated intervention in 2011, assigning treated status to companies two years before the actual implementation of the program. If the identification strategy were confounded by preexisting differences between treated and control firms, one would expect a spurious treatment effect to emerge. The estimated coefficient for this placebo treatment is -0.011 and statistically insignificant, indicating that there are no differences before treatment in revenue trends. This provides strong evidence that baseline estimates are not driven by anticipation effects or unobserved time-varying shocks correlated with treatment status.

Table 7: Robustness Checks

	<i>Dependent variable: Revenue</i>		
	Main	Balanced Panel	Placebo (2011)
	(1)	(2)	(3)
Treated $\times$ Post	0.376*** (0.028)	0.506*** (0.078)	
Fake Treated $\times$ Post			−0.011 (0.017)
Observations	52,743	6,014	16,082
R ²	0.078	0.124	0.0001
Adjusted R ²	0.069	0.098	−0.033
F Statistic	4,396.2*** (df = 1)	824.5*** (df = 1)	1.8 (df = 1)

Note: *p<0.1; **p<0.05; ***p<0.01. RSE clustered at the firm level.

4.4 Regression Discontinuity Analysis

To complement the Difference-in-Differences analysis and provide additional causal evidence, we implemented a sharp regression discontinuity (RD) design that takes advantage of the eligibility threshold of 100-employees. Firms with baseline employment just below the threshold (eligible) were compared to firms just above the threshold (ineligible), isolating the local effect of program eligibility on firm results.

Table 8 presents the RD estimate of the RD. The estimated effect of treatment at the cutoff is 0.183, which corresponds to an **approximately 18.3% increase in revenue** for firms immediately below the eligibility threshold relative to those just above it. Although the standard error reported is extremely small because of the output of the RD software, the z-value and the p-value indicate that the effect is statistically significant at conventional levels. The RD bandwidth used in this estimate was 1.523 employees and the analysis includes 944 firms within this local window.

Table 8: RDD Estimate for Revenue

Estimate	StdError	z_value	p_value	N	Bandwidth
0.183	0	2.626441e+14	1	944	1.523

The corresponding RD plot (Figure 2) displays a clear discontinuity in revenue at the 100-employee threshold. Firms immediately eligible for the program exhibit higher revenue levels compared to slightly larger firms that were ineligible, visually confirming the estimated local treatment effect. This provides complementary evidence that program eligibility causally increased firm revenue in the vicinity of the threshold, consistent with the DiD results obtained from the full panel.

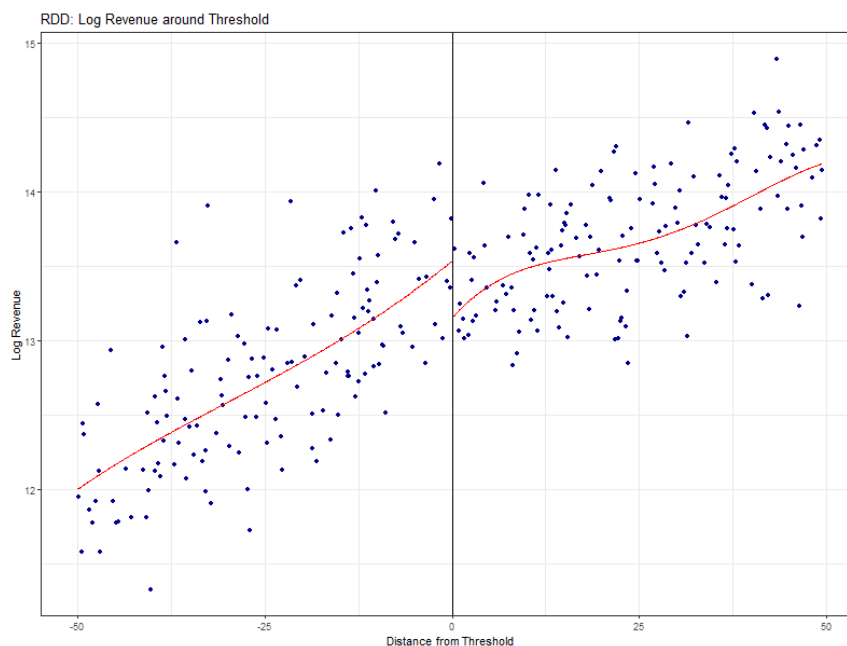


Figure 2: Regression Discontinuity Plot for Revenue around 100-Employee Threshold. The solid line represents the fitted regression on each side of the cutoff, illustrating a discrete jump in revenue at the eligibility threshold.

5 Discussion and Conclusion

This study evaluated the causal impact of a targeted training program on firm outcomes using three complementary empirical strategies: Difference-in-Differences (DiD) with two-way fixed effects, robustness checks including a placebo and balanced panel, and a Regression Discontinuity (RD) design exploiting the 100-employee eligibility threshold.

The DiD estimates indicate that firms participating in the training program experienced a **approximately 37.6% increase in revenue**, a **21.8% increase in sales**, and a **37.9% increase in revenue per worker** relative to control firms over the post-treatment period. Employment, by contrast, did not exhibit a significant increase, suggesting that the program's effects were concentrated on operational efficiency and financial outcomes rather than on workforce expansion. Robustness analyses confirm these patterns: the effect remains significant in a balanced panel of firms and is absent in the placebo test for the pre-treatment year 2011, demonstrating that the observed impacts are not driven by pre-existing trends or spurious correlations.

The RD analysis provides complementary evidence at the eligibility threshold. Firms just below the 100-employee cutoff experienced an **approximately 18.3% increase in revenue** relative to slightly larger, ineligible firms, with the discontinuity clearly visible in the corresponding RD plot. This localized effect reinforces the causal interpretation of the program's impact, particularly for smaller firms that met the eligibility criterion, and highlights the importance of the threshold in guiding program targeting.

Together, the results suggest that the training program was effective in enhancing firm financial performance, particularly revenue generation and efficiency, while the effect on employment was limited. These findings are consistent across multiple identification strategies and highlight that the program's benefits are concentrated in operational and productivity dimensions, rather than labor expansion. The study further demonstrates the value of combining DiD and RD approaches in evaluating policy interventions: DiD captures average treatment effects across the full panel, while RD isolates local causal impacts around the eligibility threshold, offering a nuanced understanding of program effectiveness.

From a policy perspective, these findings underscore that targeted training interventions can substantially improve firm performance among eligible small and medium-sized enterprises. They also suggest that eligibility criteria based on firm size are effective in concentrating program benefits among firms most likely to gain from capacity-building support. Future research could explore mechanisms such as adoption of new practices, process improvements, or managerial capability enhancements that drive the observed increases in revenue and productivity, as well as the potential long-term effects on employment and firm growth.

Bibliography

- Bloom, N., Eifert, B., Mahajan, A., McKenzie, D. and Roberts, J. (2013), 'Does management matter? evidence from india', *The Quarterly Journal of Economics* **128**(1), 1–51.
- Bruhn, M., Karlan, D. and Schoar, A. (2018), 'The impact of consulting services on small and medium enterprises: Evidence from a randomized trial in mexico', *Journal of Political Economy* **126**(2), 635–687.
- McKenzie, D. and Woodruff, C. (2014), 'What are we learning from business training and entrepreneurship evaluations around the developing world?', *The World Bank Research Observer* **29**(1), 48–82.